

Scintillations



"Hmmm... Sounds grave, very grave,
We'll know more after the autopsy!"

SOLOV - PETCT

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Case 27. How to avert false negative FNAC with the help of PET scan?

Contributed by : Dr. Sonawane, Dr. Devarti Khurjekar, Dr. Ritu Bhandari, Dr. Shrikant Solav

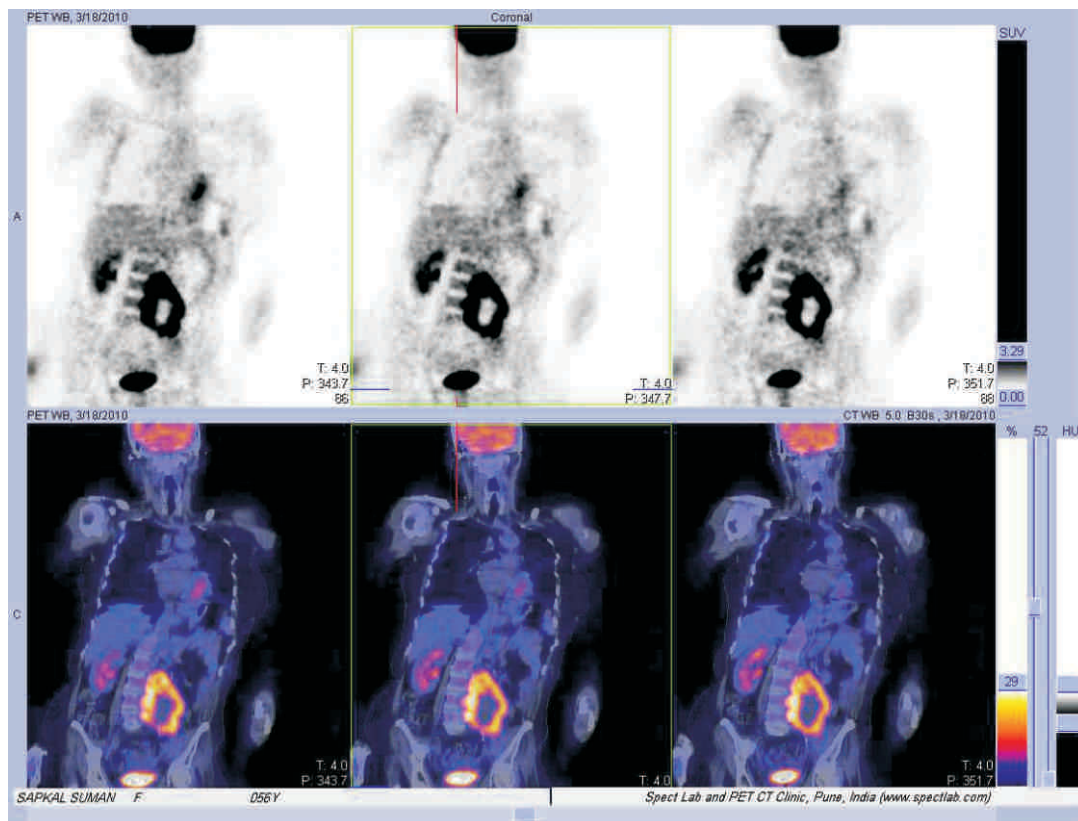


Fig. 27 A

This is a story of a 56 years old lady who was diagnosed as carcinoma of endometrium and underwent hysterectomy, bilateral pelvic lymph node dissection in 2009. She developed flank pain. An ultrasonographic examination revealed bulky heterogeneous left psoas muscle. A differential diagnosis of psoas abscess, large lymph nodal mass was made. Biopsy from the lesion revealed dense inflammatory cells.

Keeping in view the possibility of false negative FNAC, she was subjected to whole body PET CT examination. The study revealed a large metabolically active lesion in the bulky left psoas with central necrosis (please note the large central cold area Fig 27 A).

Another metabolically active pulmonary nodule was noticed in the left lobe (Fig 27 B).

The scan findings were quite suggestive of metastasis which was confirmed histologically.

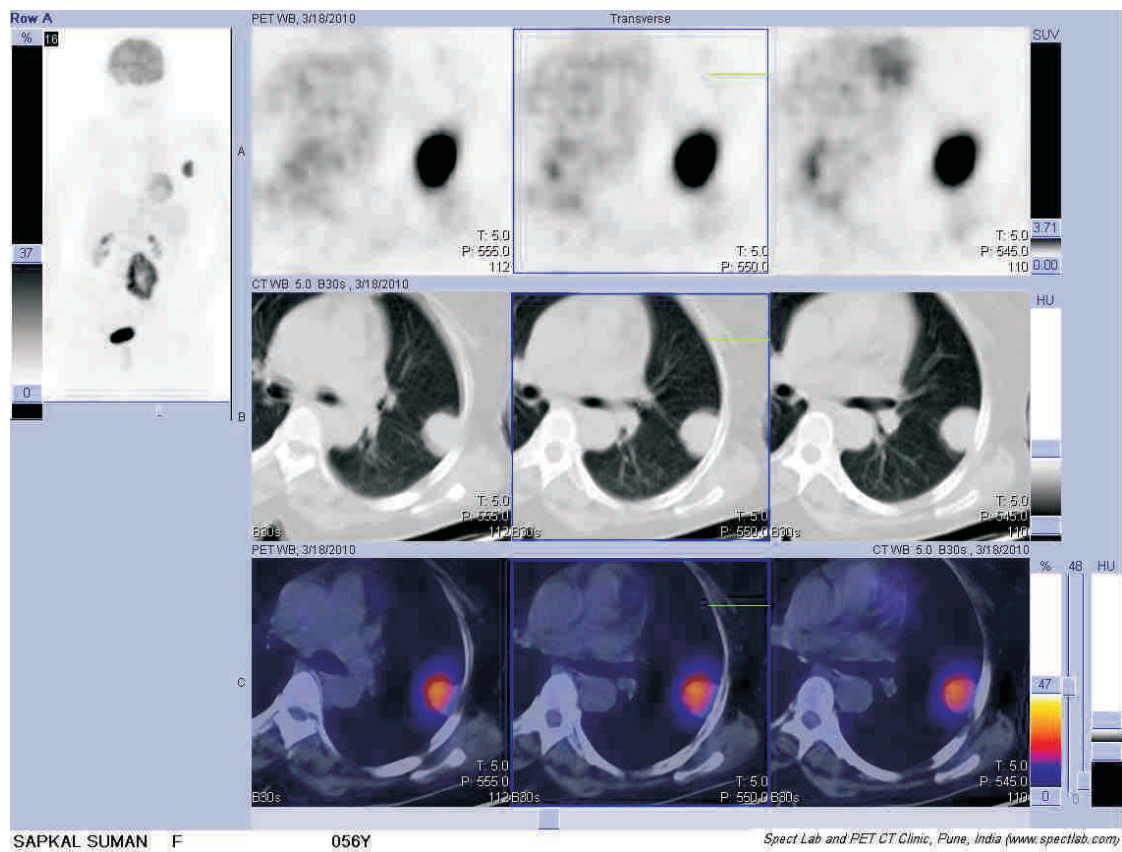


Fig. 27 B

Moral of the story- PET scan identifies the correct areas of metabolically active viable tumor tissue thus averting the possibility of false negative result that occurred in this patient because of large central area of necrosis.

Case 28. How PET scan aids in making the diagnosis of Lymphoma:

Contributed by : Dr. A A Ranade, Dr. Deepa Joshi, Dr. Devarti Khurjekar, Dr. Ritu Bhandari, Dr. Shrikant Solav



Fig. 28 A

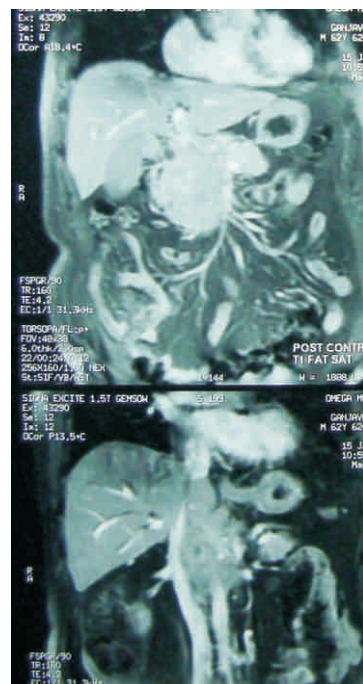


Fig. 28 B

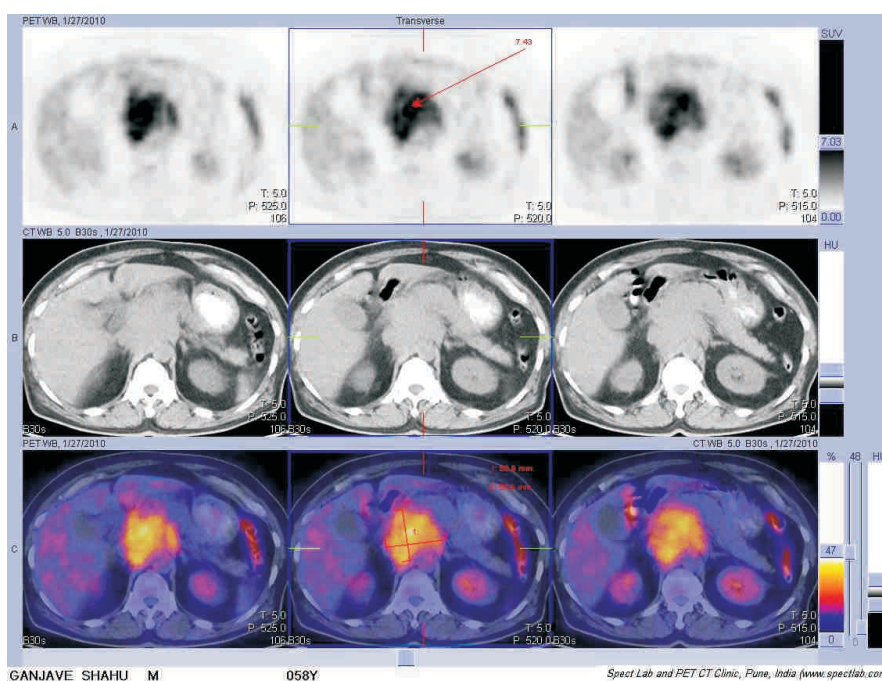
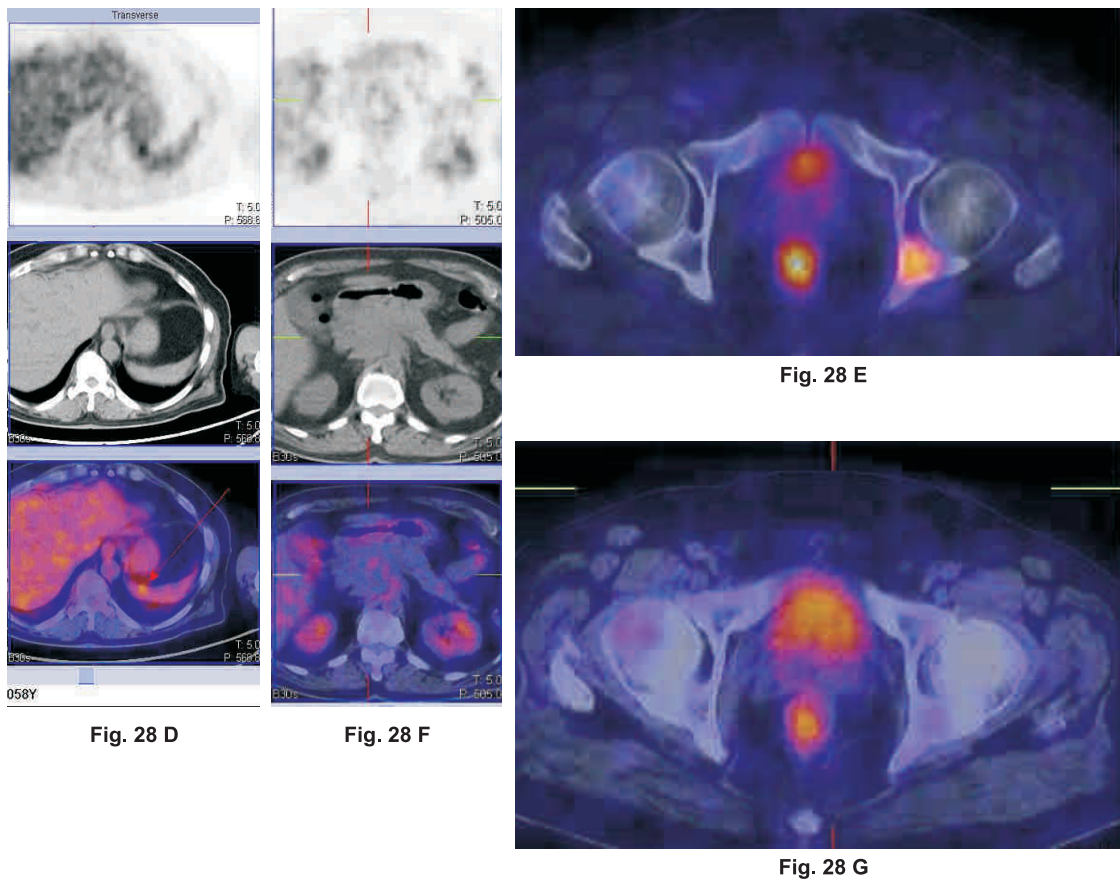


Fig. 28 C



This is the story of 58 years old gentleman who had longstanding diabetes. He complained of loss of appetite for which an ultrasound examination was requested. The study revealed a hypoechoic mass in the region of head of Pancreas displacing the portal vein anteriorly (Fig 28A), small cystic lesion within the mass, distended GB, CBD=8.9mms. His liver functions were deranged. (Bilirubin 2.27mg/dl, ALT-226 AST-129 IU/L,alkaline phosphatase-168 (N<130)).

MRI revealed large enhancing retroperitoneal mass encasing multiple retroperitoneal vessels, displacing the portal vein and duodenum and infiltrating the head of pancreas (Fig 28B). There was marrow signal abnormality in D12 L1.

Endosonographic FNAC was negative for malignant cells but CD3 and CD20 positive.

In view of the above the diagnosis was still uncertain. Hence a PET scan was ordered.

PET scan revealed a large metabolically active lymph nodal mass in the peripancreatic region (28C), peritoneal nodules, splenic nodule (28D) and skeletal lesions (Fig 28E) highly suggestive of Lymphoma.

In view of the above findings, he was offered treated for Lymphoma and a follow up PET scan post three cycles of chemotherapy showed remarkable resolution of lesions. Please note ametabolic retroperitoneal lesion as well as disappearance of left acetabular lesion. (Fig 28 F and 28G)

Case 29. Vague rib cage pain: Bone scan gives diagnostic clue:

Contributed by : Dr. Ashwini Joshi, Dr. Ritu Bhandari, Dr. Shrikant Solav

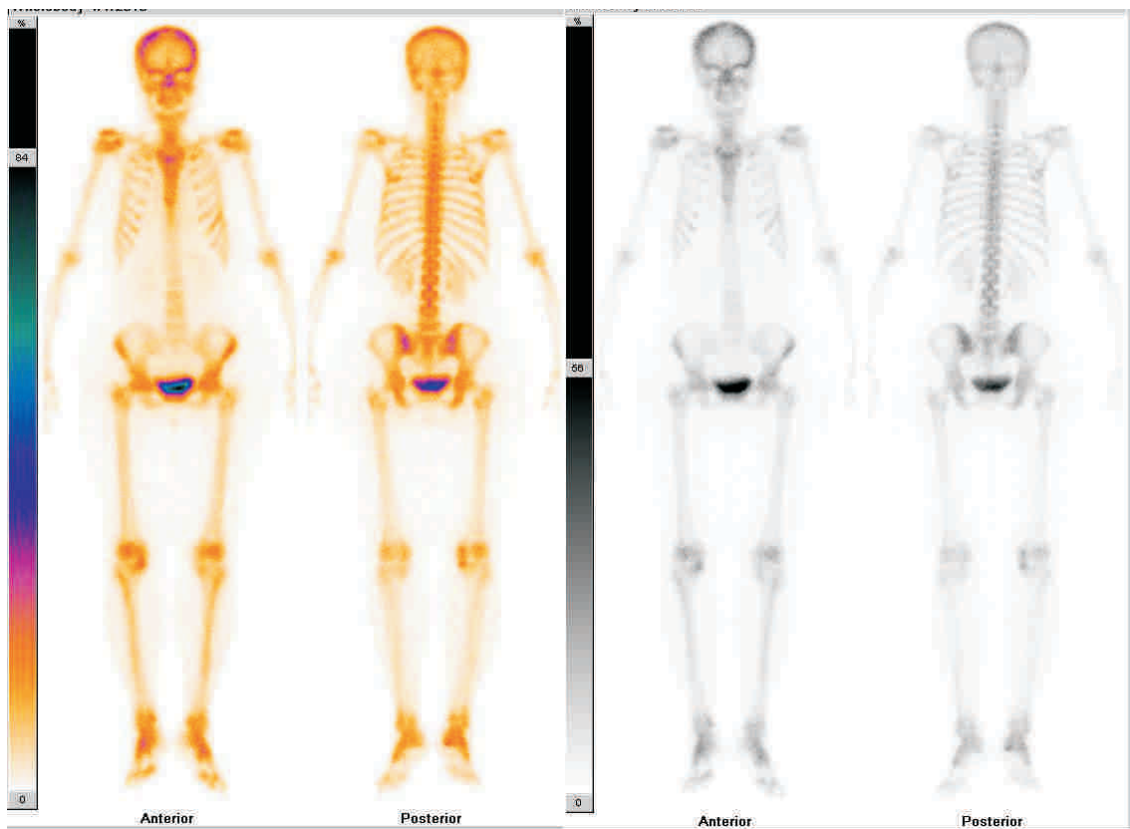


FIG 29 A

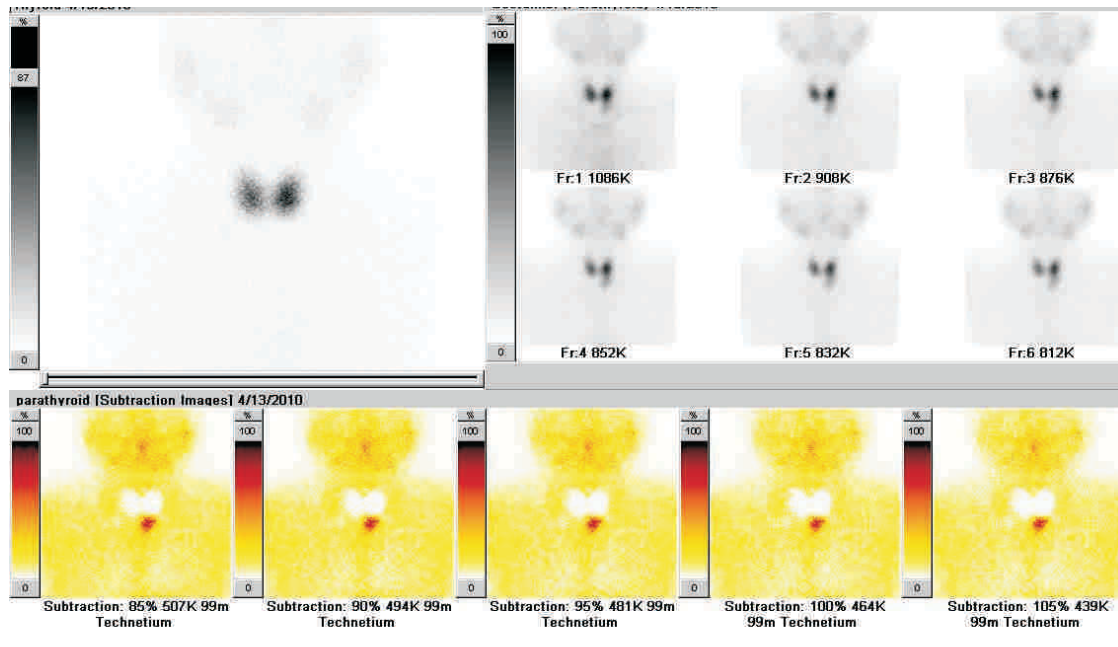


Fig 29 B

This 56 years lady presented with pain in the rib cage. The pain was ill localized. Plain radiograph of the chest was normal. Whole body bone scan was ordered to localize the site of abnormality. The study (Fig 29A) revealed diffuse increased radiotracer localization in the entire axial and appendicular skeleton. Hence a working diagnosis of Metabolic bone disease was made and blood biochemistry was requested. Subsequently the blood investigations were as follows: Serum calcium=11.27mg/dl (raised), phosphorous=2.06mg/dl (low), alkaline phosphatase=484IU/L, PTH=904pg/ml (raised). Nuclear parathyroid scan revealed well defined Adenoma in the lower pole of left lobe of thyroid gland (Fig 29B)

6th Radiation Safety Committee Meeting held on May 16th 2010



Standing (Left to Right) : Mr. Vivek Vankar, Mr. Shankar Magar, Mr. Vikas Shinde, Mr. Supriyakumar Gosh, Mr. Sanjay Bagale, Mr. Ranjit Mahajan, Mr. Appu Rai, Mr. Vasant Susar, Mr. Ganesh Chandramouli, Mr. Parag Deshmukh, Mr. P.K. Gaur, Mr. Kalyan Wable, Dr. Mrs. Ritu Bhandari, Mr. Mahesh, Miss Deepanjali Gawade, Mrs. Sangita Kamble, Mrs. Anita Pokale, Mr. Suresh Inkar.

Sitting (Left to Right) : Dr. Shrikant V. Solav, Col. Dr. Indranil Sinha, Mr. S.V. Bapat, Dr. P.S. Soni, Dr. Satbir Sachdev, Dr. N. Sivaprasad

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